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Roll No.

TPE-611

**B. TECH. (PE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

ALTERNATE ENERGY RESOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss briefly the following non-conventional energy sources in terms of latest trends and global market scenario : (CO1, CO2, CO3, CO6)
 - (i) Solar Energy
 - (ii) Wind Energy
 - (iii) Hydropower Energy
 - (iv) Biomass
- (b) What are conventional and non-conventional energy sources ? Discuss the merits and demerits of various renewable technologies developed in India. (CO1, CO2, CO3, CO6)

P. T. O.

- (c) Enumerate the criteria based on which energy can be classified. Give their availability, relative merits, demerits and their classification.
(CO1, CO2, CO3, CO6)
2. (a) Classify Wind Mills. Compare horizontal and vertical axis wind turbines.
(CO4)
- (b) Enumerate and explain *four* methods of wind recording accepted by World Meteorological Organization.
(CO4)
- (c) What are the main factors which govern the selection of site for a proposed wind turbine generating system ?
(CO4)
3. (a) Explain briefly the densification process of biomass conversion. Differentiate between "Combustion, Incineration and Pyrolysis" of biomass.
(CO5)
- (b) Classify biogas digesters. Explain briefly the components of a biogas plant.
(CO5)
- (c) Compare thermochemical and biochemical conversion process of biomass. Explain briefly the methods used to obtain energy from biomass.
(CO5)
4. (a) Classify solar collectors. State the advantages, disadvantages and applications of concentrating collectors.
(CO4)
- (b) Compare basic features among "Pyranometer, Pyrliometer and Sunshine recorder."
(CO4)
- (c) Describe the classification of solar cells based on the type of active material used. How are electron-hole pairs generated when solar radiation is directed on a semiconductor material ?
(CO4)

(3)

5. (a) Explain various types of turbines considered for use in micro-hydro resources and various types of controls available in turbines. (CO4)
- (b) With the help of a neat diagram, explain the layout of a typical micro-hydro-plant. (CO4)
- (c) Compare the relative advantages and disadvantages of turbines used in Small Hydro Power Systems (SHPs). (CO4)

Rashishtha
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Roll No.

TPE-605

**B. TECH. (PE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

PETROLEUM PRODUCTION ENGINEERING—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe methods used for removal of salt from sea water. (CO1, CO2)
- (b) Explain produce water treatment, primary and secondary process used for water treatment. (CO1, CO2)
- (c) Write short notes on the following : (CO1, CO2)
 - (i) Compliant Tower
 - (ii) Live and Drilling Load
 - (iii) Articulated Tower

P. T. O.

2. (a) Explain, what do you understand by perforation, types of guns and explosive used in perforation. (CO1, CO2, CO5)
- (b) Define well activation. Explain the concept of swabbing and why swabbing is necessary process in well activation. (CO1, CO2, CO5)
- (c) What are gathering pipelines ? Explain, what are production riser ? (CO1, CO2, CO5)
3. (a) Explain the concept of reservoir deliverability and illustrate all the several factors on which reservoir deliverability depends. (CO2, CO3, CO5)
- (b) Write a short note on Well Bore Performance and IPR. (CO2, CO3, CO5)
- (c) Explain the following : (CO2, CO3, CO5)
- (i) Gas Well Deliverability
 - (ii) Choke Performance
 - (iii) Oil Well Performance
4. (a) Describe Sucker Rod Pump and Electrical Submersible Pumps. (CO1, CO2)
- (b) With the help of a diagram, explain intermittent gas flow system and progressive cavity pumps system. (CO1, CO2)
- (c) Write the advantages and disadvantages of hydraulic jet pumps and gas lift system. (CO1, CO2)

(3)

5. (a) Differentiate between Gas cap drive, Gravity drainage drive and Combination drive. (CO2, CO3, CO4)
- (b) What do you understand by secondary recovery. Also explain water injection and gas injection. (CO2, CO3, CO4)
- (c) Describe enhance oil recovery and write all the methods off enhance oil recovery and explain any *one* method. (CO2, CO3, CO4)

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mohd 9935205600
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Roll No.

TPE-604

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
PETROLEUM DRILLING ENGINEERING—II**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define directional drilling and deviation control. Discuss the four different situations in which directional drilling are necessary. (CO1)
- (b) What is directional drilling ? List and explain the three specific parameters to be considered in planning direction well path by suitable sketch. (CO2)
- (c) Describe the various components of a PDM. (CO1)

P. T. O.

2. (a) What are the applications of horizontal well ? What factor should be considered when designing a horizontal well ? Discuss. (CO3)

- (b) Given the following well data : (CO3)

Build curve radius (R) = 1042 ft

Horizontal section = 1500 ft

WOB = 30,000 lbf

Mud weight = 9.2 ppg

BHA is made up mostly of HWDP

HWDP = 5" , 50 lb/ft, OD of tool joint is 6.5"

Calculate torque while rotating off bottom.

- (c) What is a multilateral well ? Explain different types of multilateral well with the help of a diagram. List the most important considerations in planning a multilateral well. (CO3)

3. (a) What is differential sticking ? Describe the method for freeing differential stuck. (CO4)

- (b) Determine the magnitude of the differential sticking force across a permeable zone of 30 ft in thickness using the following data : (CO4)

Differential pressure = 500 psi

Area of contact is 20% of effective drill pipe perimeter

Filter cake = 1/2 in (12.7 mm); Friction factor = 0.1.

Drill pipe OD : 5"

- (c) Describe the causes of lost circulation and curing methods. (CO4)

4. (a) List and discuss the factors influencing casing design. (CO5)

- (b) Name the *three* types of load cases used in detailed casing design.

Discuss burst criteria. (CO5)

- (c) Calculate burst pressures for the following well : (CO5)

CSD = 15000 ft

Pf = 8500 psi

Packer fluid = 15 pp

5. (a) List the components of drillstring. Discuss the three parameters to be considered in drill pipe selection in drill string design. (CO6)

- (b) Describe the API system of drill pipe classification. Describe, how washouts develop and detected. (CO6)

P. T. O.

(4)

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- (c) Determine the size and number of drill collars required to provide a weight-on-bit of 55,000 lbf assuming : (CO6)

Hole deviation = 0°

Mud density = 12 ppg

Assume that the available drill collar size is OD/ID, 9/3 inch

The weight per foot for this size is 192 lb/ft.

Drillcollars come in 30 ft lengths.

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TPE-603

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

END SEMESTER EXAMINATION, 2022

NATURAL GAS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the following :

(CO1, CO2)

(i) Retrograde condensation

(ii) Critical point

(iii) Bubble point

(iv) Dew point

(v) Gas hydrates

P. T. O.

- (b) Define the quality lines in phase diagram. Draw the PT and PV phase diagram for single component system showing the different region. What is natural gas ? Discuss all the properties with mathematical expressions of natural gas. (CO1, CO2)
- (c) Calculate the gas specific gravity of natural gas in the reservoir having the following composition : (CO3)

Composition	Mole Fractions
C1	0.775
C2	0.083
C3	0.021
CO2	0.030
H2S	0.020

The reservoir is having at initial condition of $P_i = 3500$ psia and 185°F . By calculating sp. gr. of the gas also calculate T_{pc} , P_{pc} , T_{pr} , P_{pr} by using Standing and Katz correlations (take compressibility factor $Z = 0.87$ for the data given) and hence the density of natural gas. Compare the density of the natural gas in case of ideal gas.

2. (a) With the help of a diagram ? Explain the extraction of natural gas by the enhanced (CBM) techniques from CBM. Write down the usage and advantages of CBM. What do you mean by proppant ? What are the functions of proppant ? Discuss the types of proppant. (CO3, CO4)

- (b) Describe the steps involved in the extraction of shale gas. Why hydraulic fracturing needed in the production of shale gas ? Explain hydraulic fracturing. Illustrate the different theories and techniques for the hydraulic fracturing. (CO1, CO4)
- (c) How gas hydrates formed ? Discuss the various issues related to extraction of natural gas hydrates. Gas hydrates extraction requires the unconventional approach, why ? Explain the different methods in the extraction of gas hydrate. (CO1, CO2)
3. (a) Classify the Sweetening process. What are the different amines used in the gas amine based sweetening process ? Define iron-sponge sweetening. Write down the reactions involved in the iron sponge process. (CO1)
- (b) What are the informations that is to be known for the design of glycol dehydrator ? In the design of glycol dehydrator, explain glycol contractor, water rate and required glycol circulation rate along with their mathematical expression. (CO1, CO2)
- (c) What are the dehydration systems ? Discuss absorption and adsorption process in the dehydration of natural gas. (CO1, CO5)
4. (a) Define Sulfinol process with its main advantage in the natural gas sweetening. What are the criteria for the selection of different sweetening process ? (CO1, CO5)
- (b) Define recording charts. What are the *two* types of principal chart used ? Discus in detail about direct reading charts. (CO4, CO5)

- (c) What are the different methods/instruments used in the volumetric measurement of natural gas ? Discuss orifice meter for the measurement of natural gas with required equations used. (CO3, CO6)
5. (a) Discuss natural gas utilization as fertilizer and petrochemical feed stock in detail. (CO4, CO6)
- (b) Calculate the maximum quantity of gas that can be transported through 4" dia, 20 km long horizontal pipeline under the following conditions : (CO4, CO6)

Inlet pressure = 2000 psia

Maximum pressure drop = 500 psi

Average gas temp. = 100 F

Pipeline efficiency factor = 0.92

Gas gravity = 0.6

Zavg can be taken as = 0.9

Use both Panhandle B and Weymouth methods.

- (c) Define transmission factors. Tabulate and discuss the different transmission factor with different flow equations. (CO2, CO3)

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TPE-601

B. TECH. (PETROLEUM ENGINEERING)

(SIXTH SEMESTER)

END SEMESTER EXAMINATION, 2022

RESERVOIR MODELING AND SIMULATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How many types of rocks are available on the earth ? Explain sedimentary rocks and their composition and relative abundance on the earth. (CO2)

(b) What are the *three* essential properties of the rocks to be reservoir ? Explain in detail. Name the various types of physical properties of rock. (CO1)

P. T. O.

- (c) Explain various types of model required for exploration and development of oil and gas field. What are the various types of software use for building these models ? (CO4)
2. (a) Explain structural and stratigraphic traps in detail. Draw a cross-section of a petroleum system and explain its all important components. (CO3)
- (b) Explain Reservoir Modeling, Reservoir Engineering and various step of Reservoir Engineering for better field development. (CO4)
- (c) Explain the term sequential workflow in context with the modeling process. What are the essential data required for building a reservoir model ? (CO5)
3. (a) Explain drive mechanism,. Illustrate various drive mechanisms of petroleum system with figure in detail. (CO2)
- (b) How does packing and sorting affect the porosity and permeability ? Define with figure cubical packing, orthorhombic packing and rhombohedra-packing of spheres along with their porosity values. (CO4)
- (c) Design a suitable workflow for making a reservoir model and explain its various steps in detail. (CO3)
4. (a) Explain Primary, Secondary, Total and Effective Porosity with example, permeability and oil saturation with figure. (CO1)

(3)

- (b) What is traps ? Illustrate different types of traps by drawing a neat figure. How does the trap mechanism take place in the petroleum system ? (CO3)
- (c) What are the uses of data import, visualization, data editing and interpretation well correlation of various sequences ? (CO5)
5. (a) Explain OWC, GOC, OSC and free water. Draw a plot of water saturation profile in an oil-water system. (CO3)
- (b) What is permeability ? Explain Darcy's equation, How is it use to measures permeability for flowing fluid and gas system ? (CO4)
- (c) Identify the various types of data required for fault and structure modeling process. Explain in detail. (CO5)